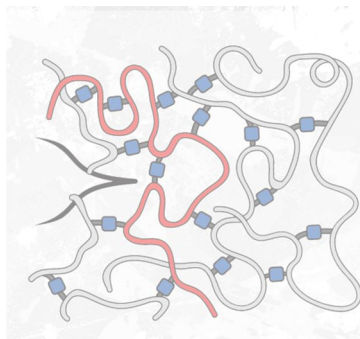
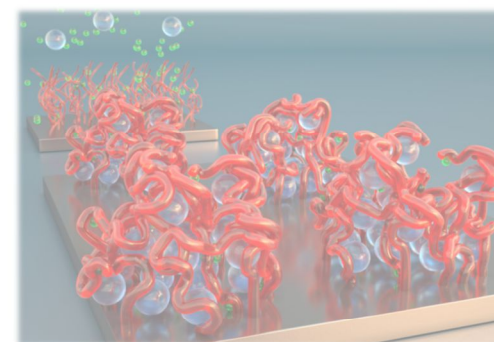




Soft Matter Systems as part of the Advanced Materials module

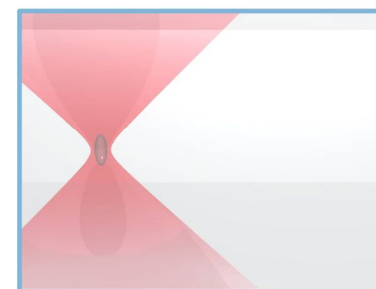
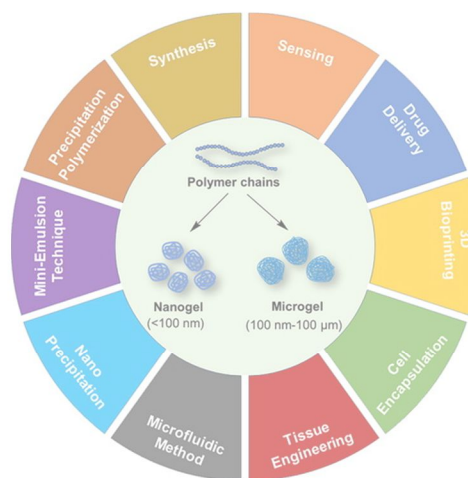


Prof. Dr Guillaume Delaittre

Chair of Macromolecular Chemistry

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www.macrochem.hhu.de



Lecture

Summer Semester 2026

Soft Matter Systems

Definitions of Soft Matter

JGU Mainz – Department of Statistical Physics and Soft Matter Theory

„This relatively new term refers to a broad class of systems including rubber, plastic, paper, foams, soap solutions, liquid crystals, blood, biomembranes, biopolymers, and many others. Soft materials share a number of common features: energies are of the order of the thermal energy kT . Therefore, the systems respond strongly to weak external forces, and their properties are predominantly determined by entropic effects. In that sense, they resemble simple fluids. In contrast to simple fluids, however, soft materials display structure not only on the atomic scale, but also on mesoscopic scales (nano- or even micrometers).“

MIT (USA) – Department of Materials Science and Engineering

“Soft matter refers to a wide range of materials, such as polymers, gels, and foams. What sets these materials apart is they can be easily molded, shaped, or changed without requiring intense heat or complicated processes. Unlike metals or ceramics, soft matter is made up of bigger building blocks, which gives it unique chemical and physical properties.”

Soft Matter Systems

Contents

Part 1 Crosslinked Polymers

Crosslinking, Gelation, Swelling, Elasticity

Thermosets

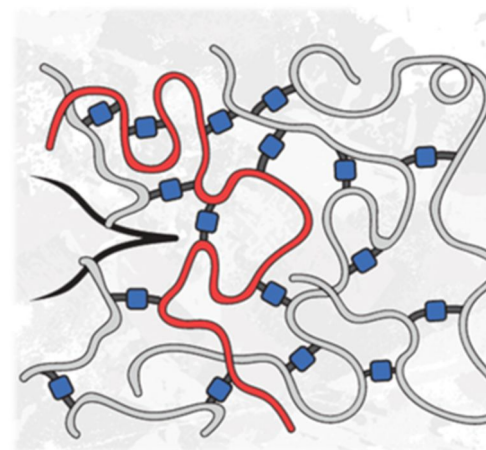
Covalent Adaptable Networks

Hydrogels

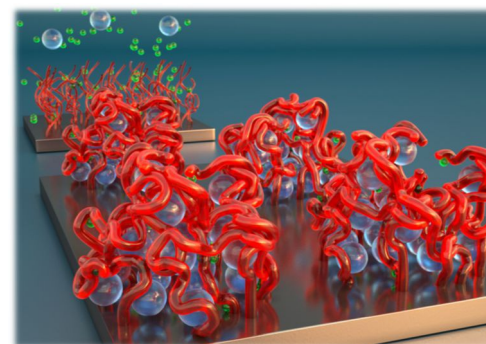
Microgels

Nanogels

Single-Chain Nanoparticles



Part 2 Polymer Brushes



What Types of Polymers are There?

Origin

Polymers often are not made of a single repeating unit.

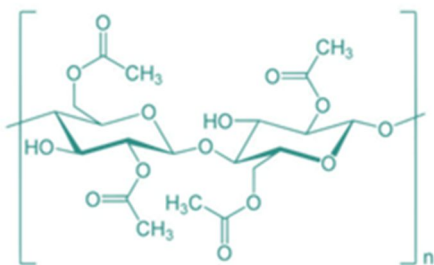
Many polymers are produced from more than one monomer: They are copolymers.

Copolymers are produced industrially and used in real-life, large-tonnage applications.

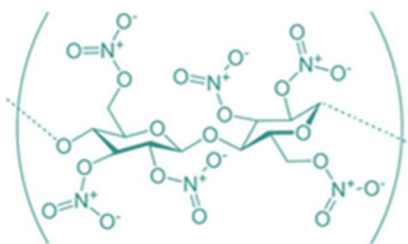
natural polymers

synthetic polymers

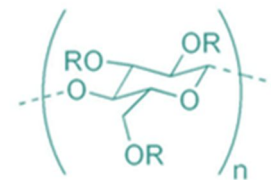
semi-synthetic polymers



Cellulose acetate



Cellulose nitrate



R = H or CH₂CO₂H

Carboxymethyl cellulose

What Types of Polymers are There?

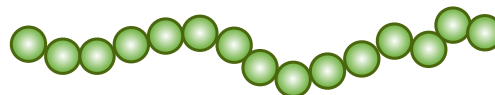
Composition and Microstructure

Polymers often are not made of a single repeating unit.

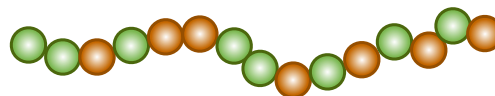
Many polymers are produced from more than one monomer: They are copolymers.

Copolymers are produced industrially and used in real-life, large-tonnage applications.

homopolymer



*copolymer
(heteropolymer)*

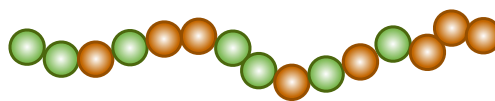


What Types of Polymers are There?

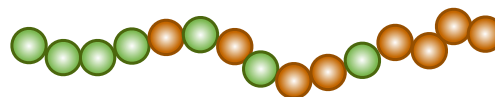
Composition and Microstructure

There are different ways of arranging the comonomers.

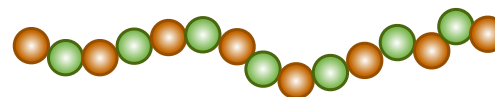
*random/statistical
copolymer*



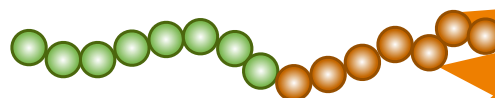
*gradient
copolymer*



*alternating
copolymer*



*block
copolymer*

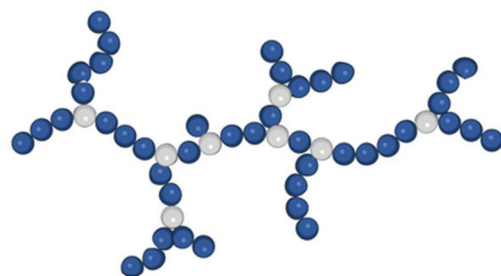


MSc
(AMC)

What Types of Polymers are There?

Architecture/Topology

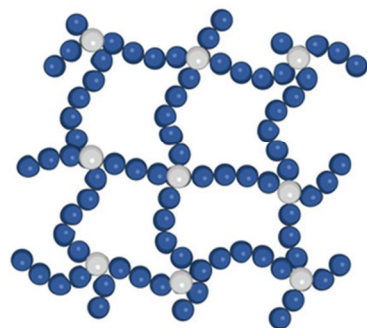
Besides, polymers often are not linear. They can contain designed or undesired branching or crosslinking points.



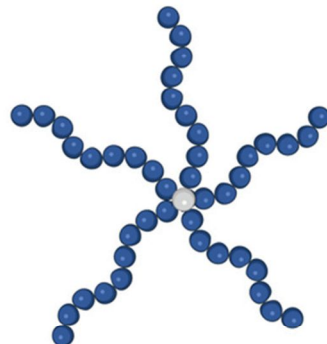
Branched polymer



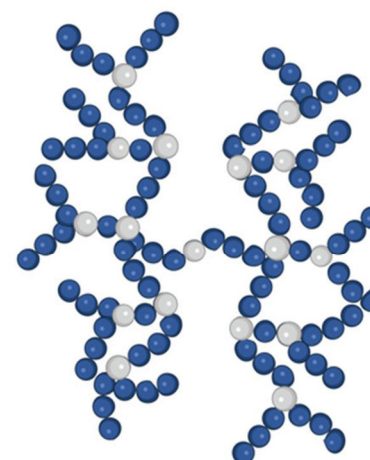
Graft polymer



Crosslinked polymer



Star polymer



Dendrimer



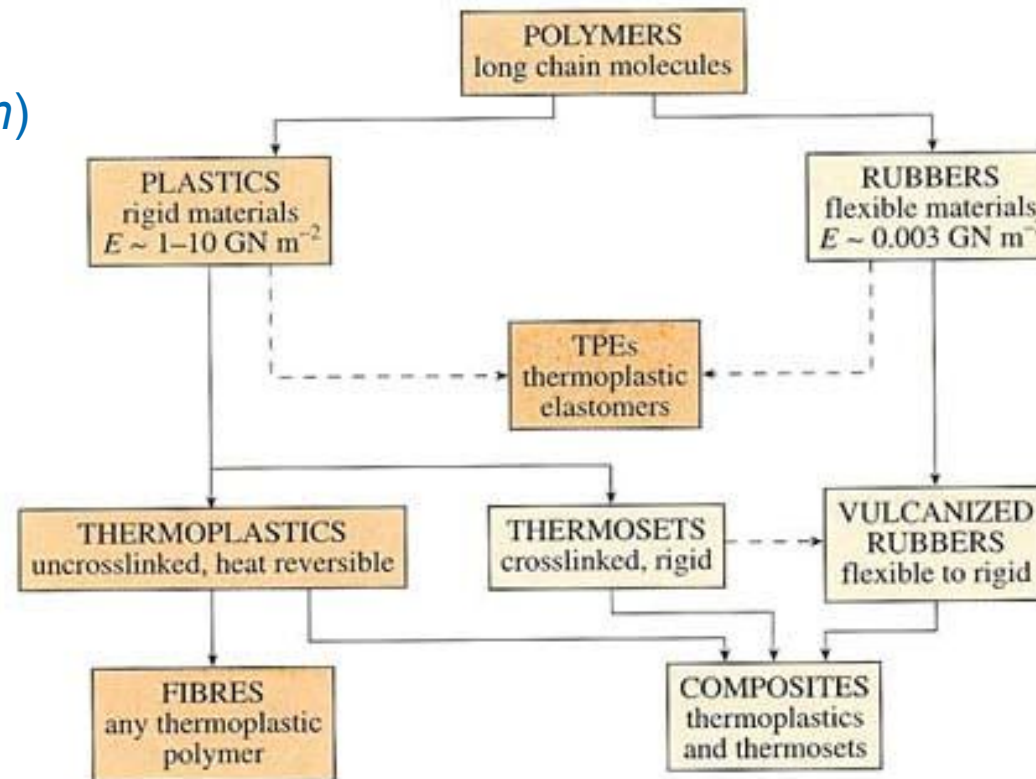
Polym. Chem. **2024**, *15*, 2800

What Types of Polymers are There?

Physical Properties

Polymers and the corresponding plastics can vary drastically in their mechanical behavior, also depending on the temperature, due to differences in involved molecular forces.

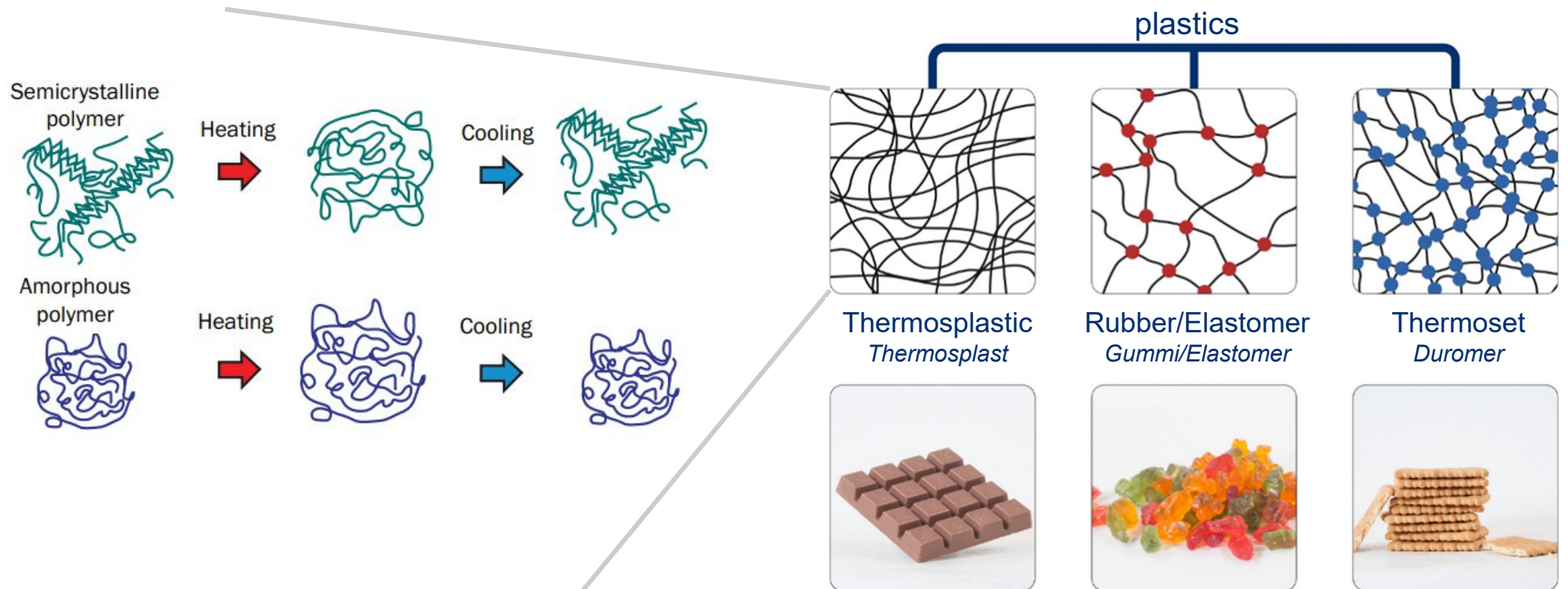
→ Plastics types
(Kunststoffgruppen)



What Types of Polymers are There?

Physical Properties

Polymers and the corresponding plastics can vary drastically in their mechanical behavior, also depending on the temperature, due to differences in involved molecular forces.

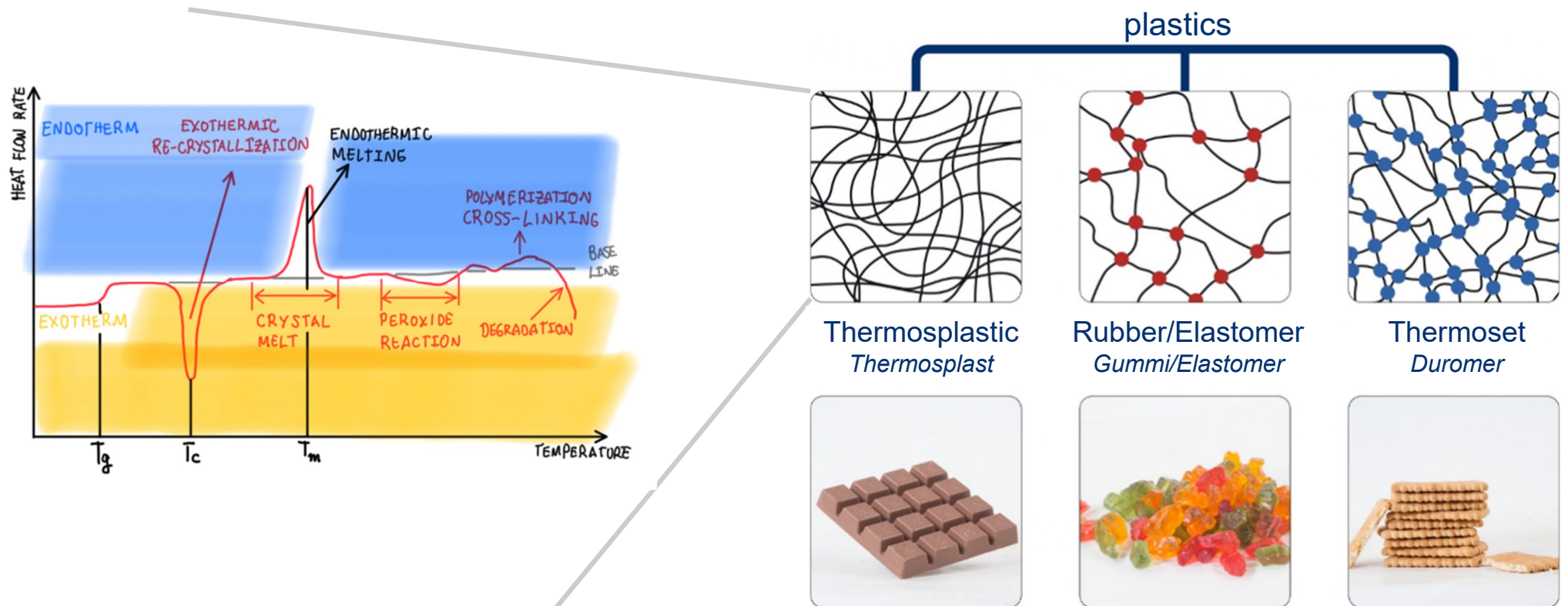


Bonten, *Kunststofftechnik* (Hanser) 2020

What Types of Polymers are There?

Physical Properties

Polymers and the corresponding plastics can vary drastically in their mechanical behavior, also depending on the temperature, due to differences in involved molecular forces.



<https://automotivepapers.com/2024/04/10/differential-scanning-calorimetry-dsc-part-1/>

Bonten, *Kunststofftechnik* (Hanser) 2020

Crosslinked Polymers

Crosslinked Polymers

Definitions

Crosslink

In chemistry and biology, a cross-link is a bond or a short sequence of bonds that links one polymer chain to another. These links may take the form of covalent bonds or ionic bonds and the polymers can be either synthetic polymers or natural polymers (such as proteins). [Wikipedia “Cross-link”]

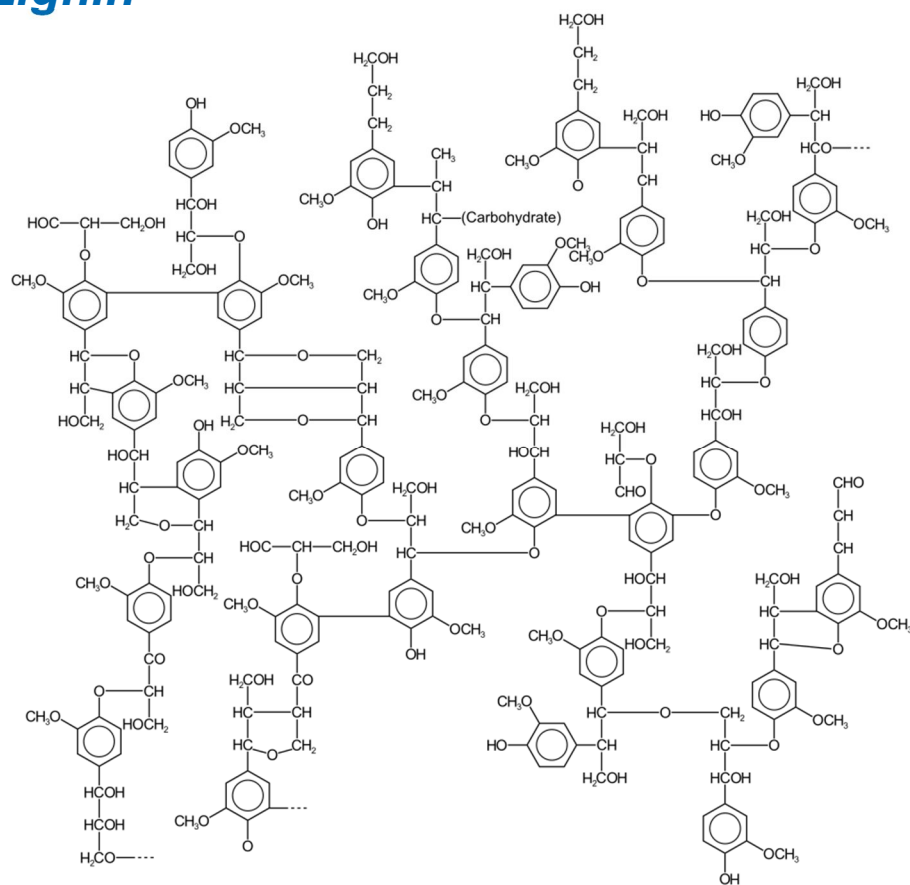
Crosslinked polymers

A crosslinked polymer is a type of polymer where individual polymer chains are connected by bonds, forming a three-dimensional network structure. This crosslinking influences the mechanical strength, elasticity, and thermal stability. They exhibit very high molecular weights and are essentially macroscopic scale molecules.

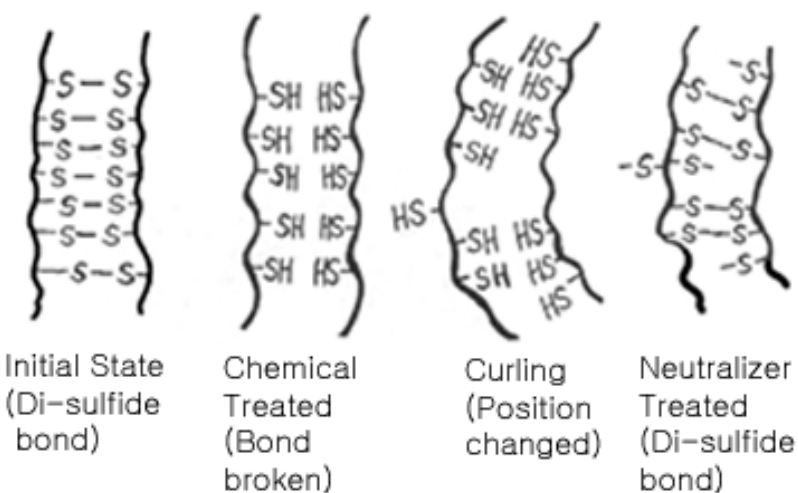
Crosslinked Polymers

Examples from Nature

Lignin



Keratin



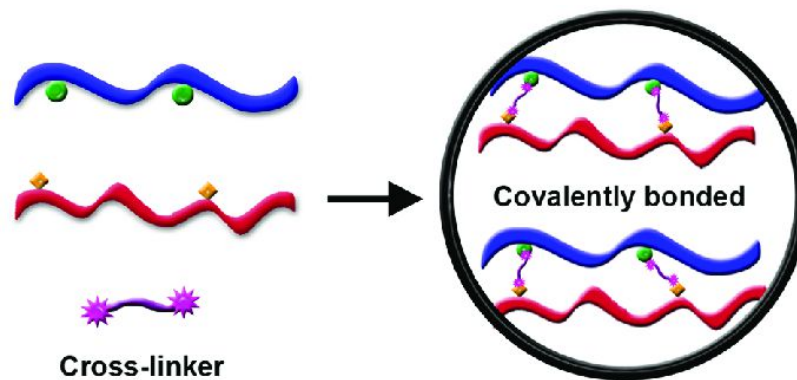
„The Chemistry Behind the Permanent Wave”

<https://evoice.ewha.ac.kr/news/articleView.html?idxno=517>

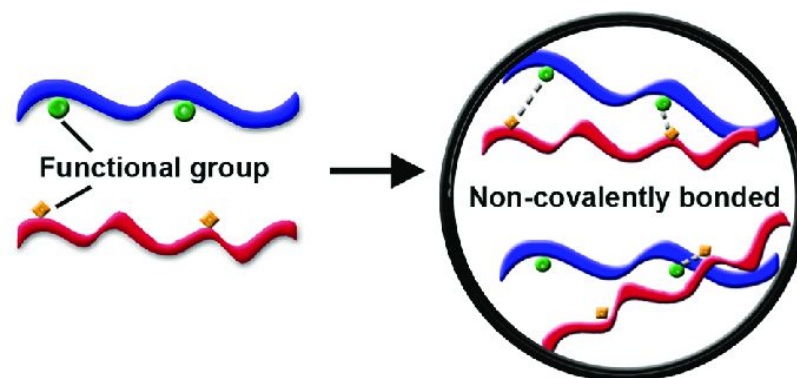
Crosslinked Polymers

Crosslinking: Covalent vs Physical (I)

Covalent crosslinking
→ permanent network



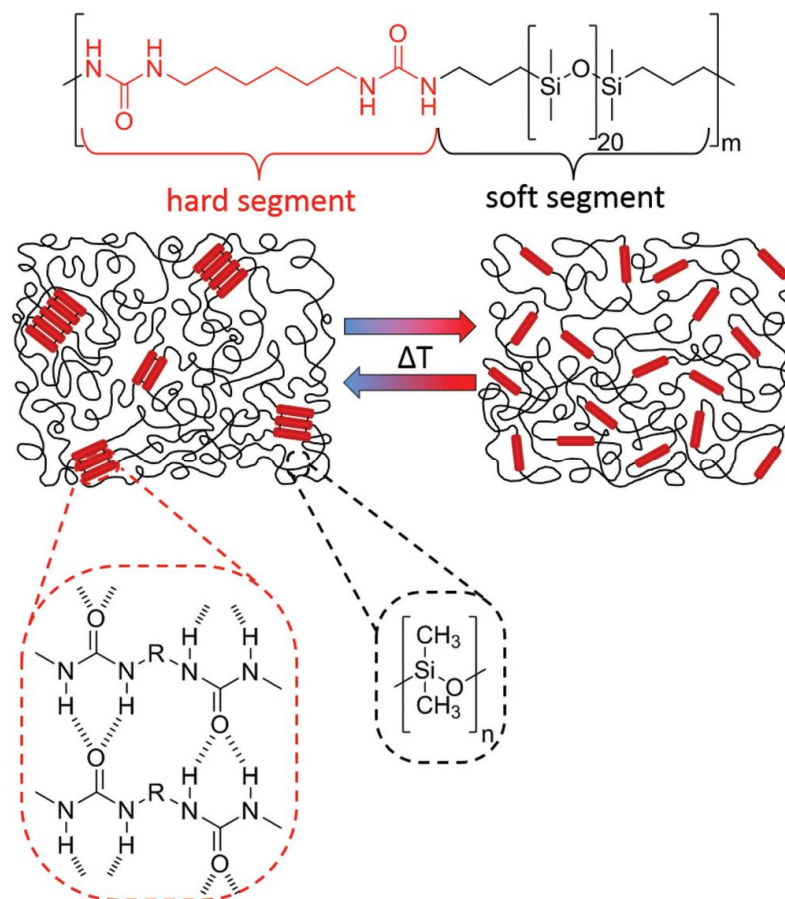
Physical crosslinking
→ reversible network



Crosslinked Polymers

Crosslinking: Covalent vs Physical (II)

Example of thermoplastic elastomers



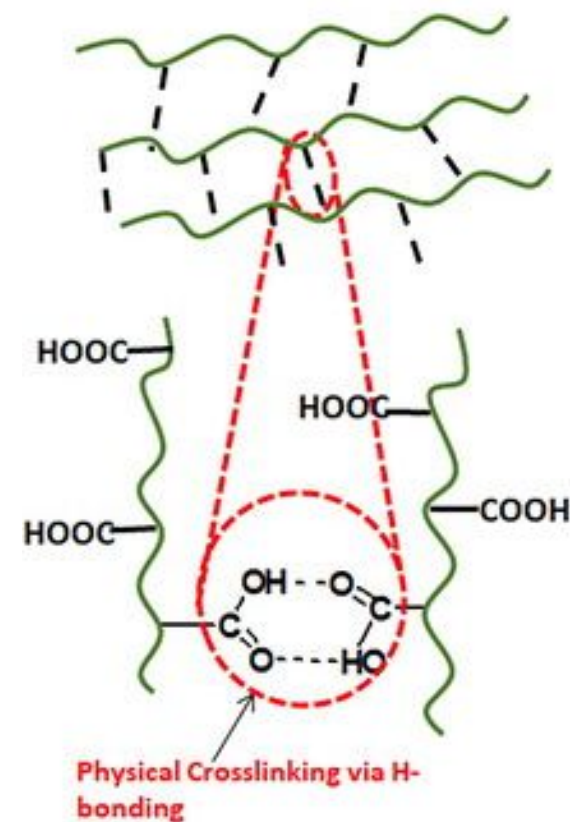
Macromol. Rapid Commun. **2018**, 39, 1800055

Crosslinked Polymers

Crosslinking: Covalent vs Physical (III)

Physical crosslinking is based on **weak interactions** and/or **self-assembly/organization**:

- π - π interactions
 - H bonding
 - electrostatic interactions
-
- crystallization
 - phase separation
 - hydrophobic effect



J. Macromol. Sci. A **2024**, 61, 265

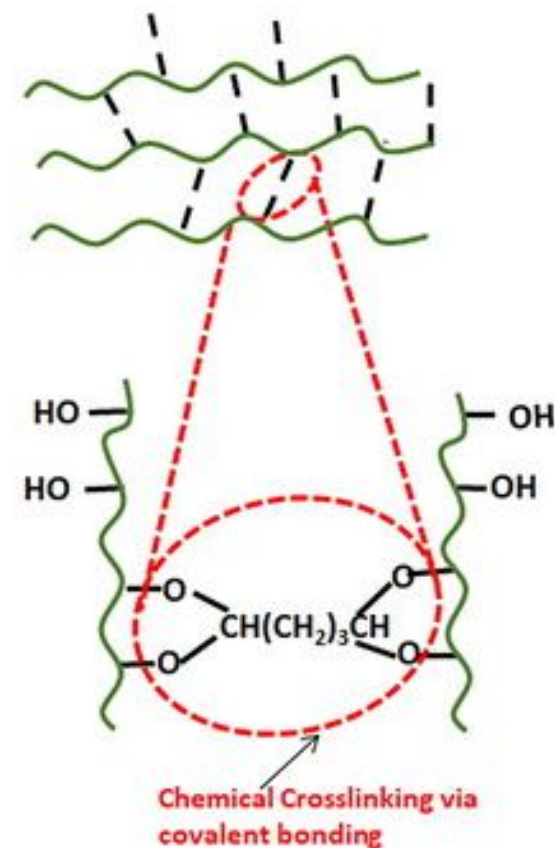
Crosslinked Polymers

Crosslinking: Covalent vs Physical (IV)

Covalent intermolecular bridging on polymer chains can occur via many different coupling strategies

either using functional groups already typically present in polymers (e.g., $-\text{C}=\text{C}-$, $-\text{OH}$, $-\text{COOH}$, $-\text{SiH}$)

or by introducing special functional comonomers with the aforementioned functionality but also with more exotic ones (e.g., $-\text{N}_3$, $-\text{C}\equiv\text{CH}$, $-\text{SH}$, $-\text{C}(\text{R})=\text{O}$).



J. Macromol. Sci. A **2024**, 61, 265

Crosslinked Polymers

Gelation & Percolation (I)

Gelation is the process where a fluid transforms into a solid-like, viscoelastic gel.

Percolation is a spreading process, leading to the sol–gel transition.

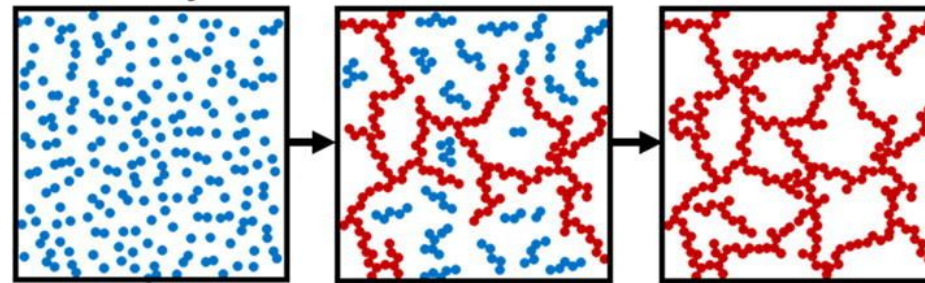


Crosslinked Polymers

Gelation & Percolation (II)

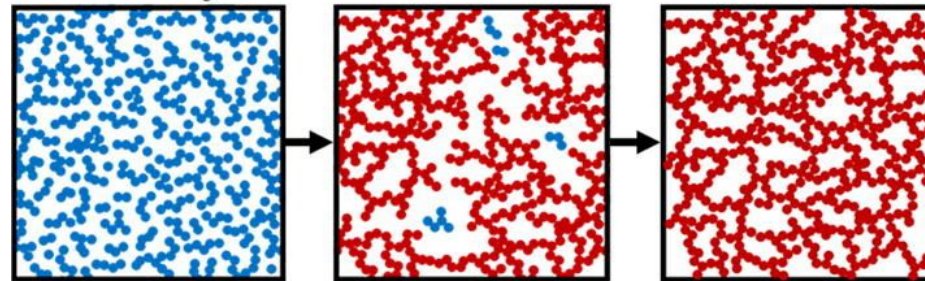
The phenomenon is known for polymers, as well as for colloids.

Dilute System



Gelation of sol clusters

Dense System



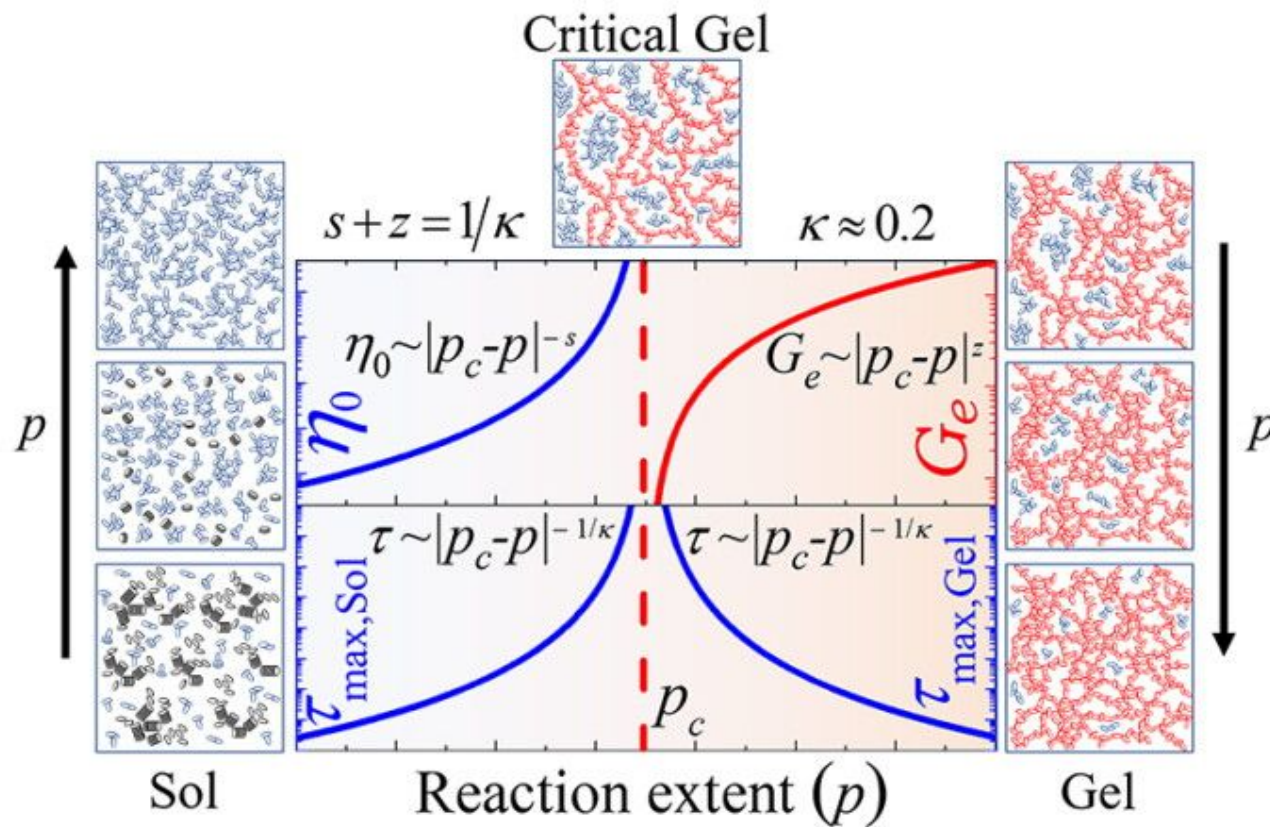
Gelation of sol clusters Aggregation of gel clusters

J. Coll. Interf. Sci. A **2019**, 550, 57

Crosslinked Polymers

Gelation & Percolation (III)

Upon gelation, the mechanical properties drastically change.

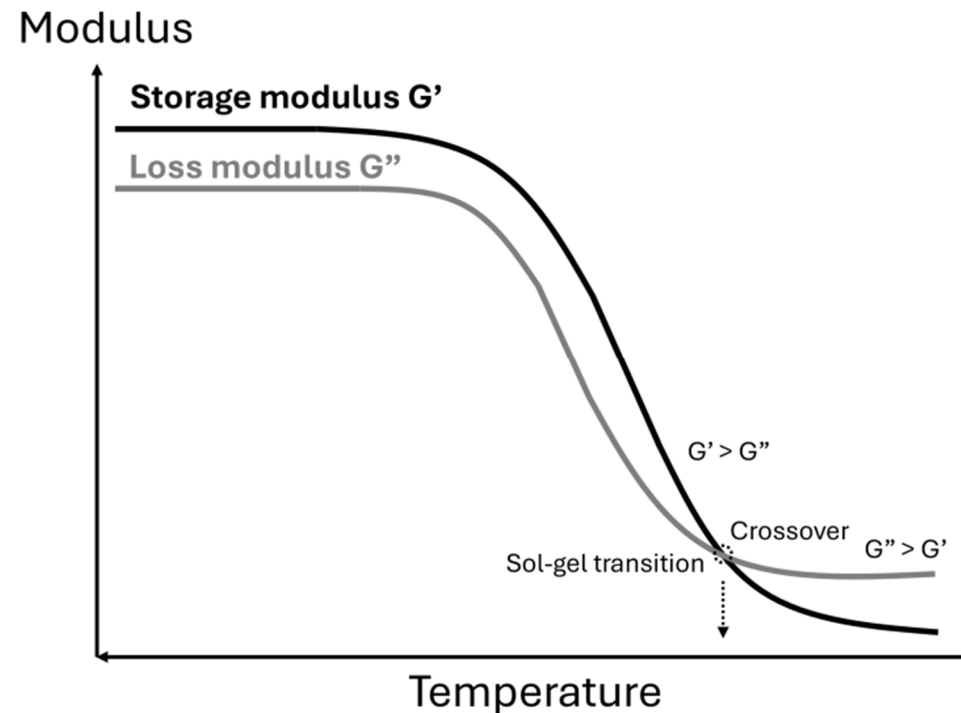


J. Rheol. **2020**, *64*, 863

Crosslinked Polymers

Sol–Gel Transition (I)

From a rheological standpoint, the sol-gel transition reflects a shift from viscous-dominated behavior (liquid-like) to elastic-dominated behavior (solid-like).

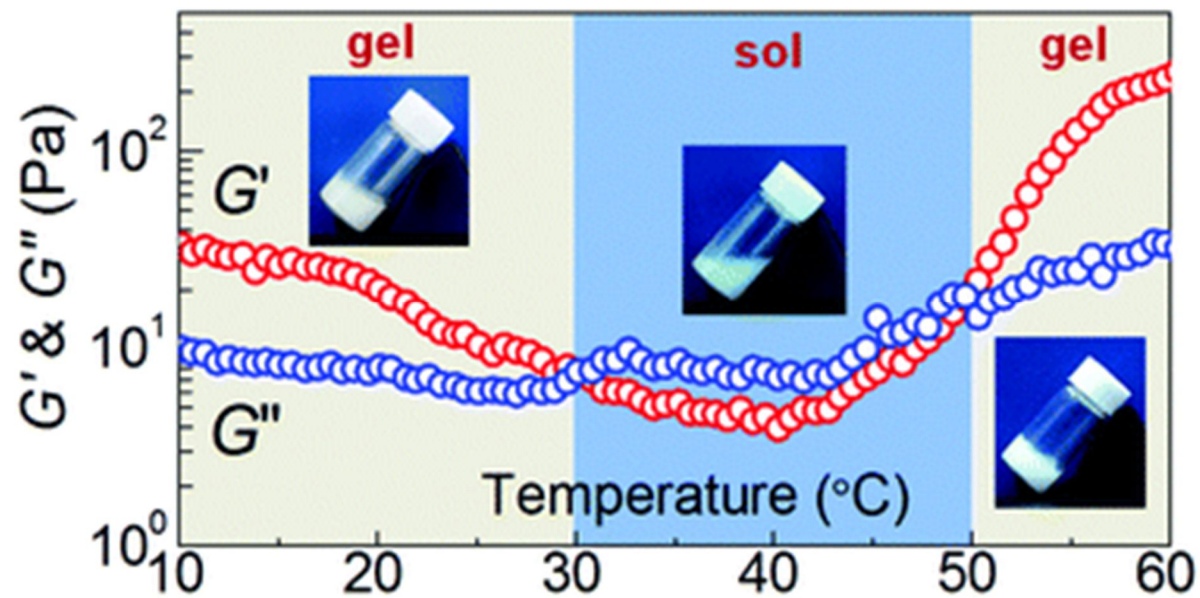


<https://rheologylabs.com/testing/material-and-ingredients/sol-gel-transition-and-gelling-behavior/>

Crosslinked Polymers

Sol–Gel Transition (II)

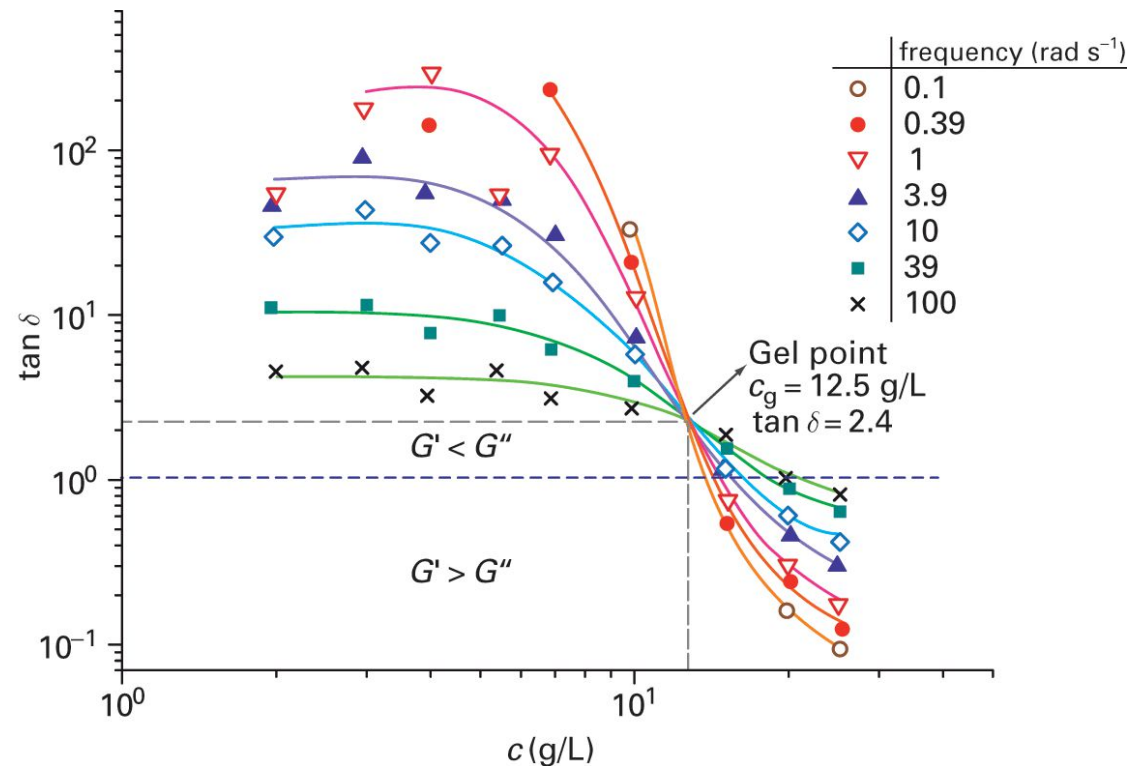
The transition is also macroscopically visible.



Crosslinked Polymers

Sol–Gel Transition (III)

The gel point is formally reached when $\tan \delta (= G''/G')$ becomes independent of the stimulation frequency in a mechanical test.

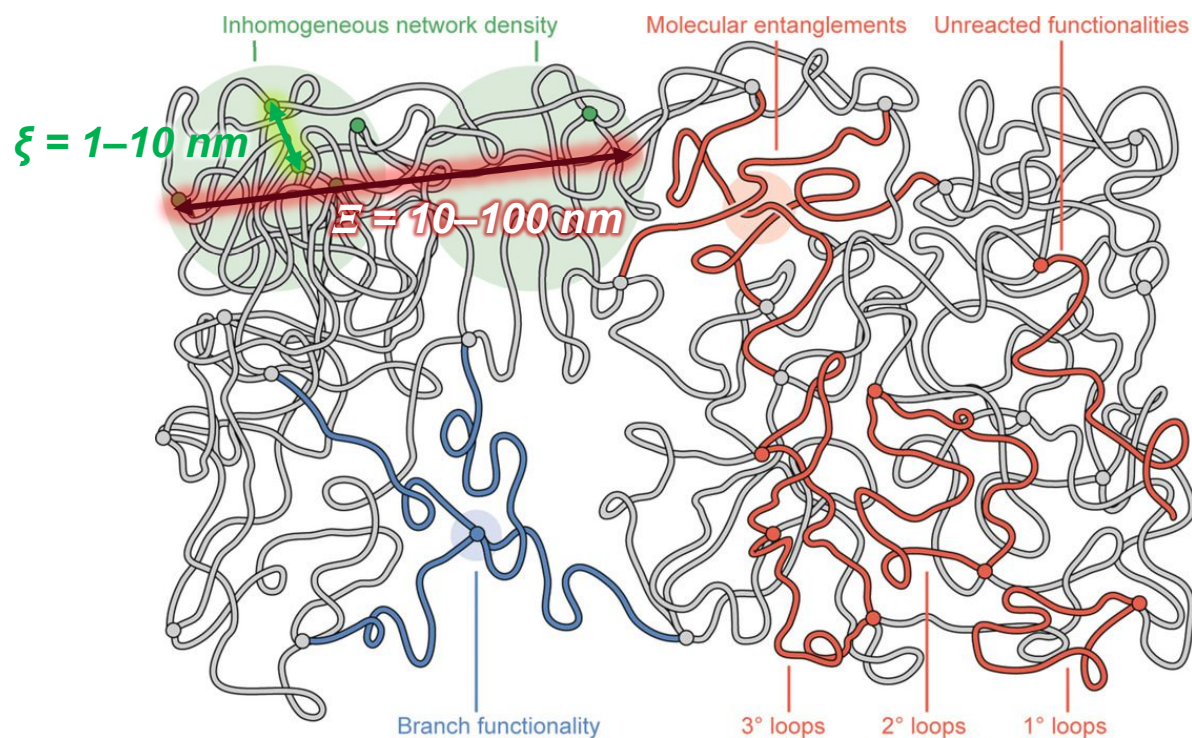


Chapter 3 – The sol-gel transition in *Physical Gels from Biological and Synthetic Polymers*, pp. 64–96

Crosslinked Polymers

Network Defects (I)

Most polymer networks, in particular those obtained by free-radical crosslinking copolymerization, exhibit a marked degree of nanostructural heterogeneity.

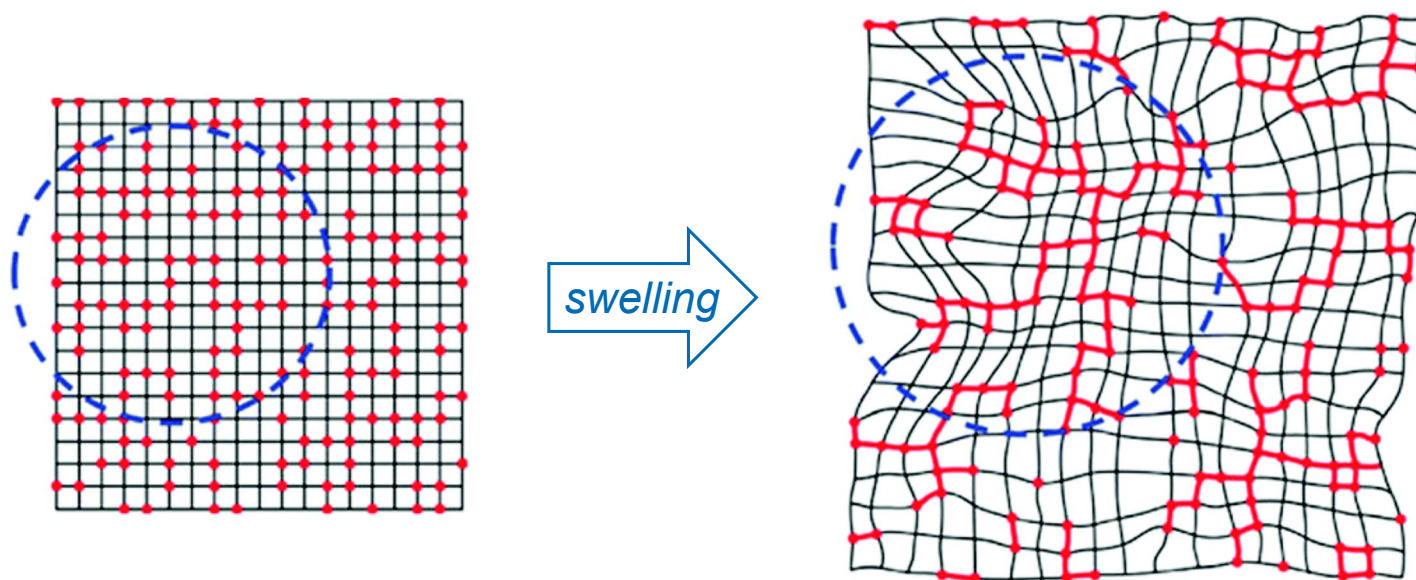


Angew. Chem. Int. Ed. **2020**, *59*, 5022

Crosslinked Polymers

Network Defects (II)

Inhomogeneity of the crosslinking density of a polymer network is also present on a further level that only appears when the polymer network is swollen in a solvent.

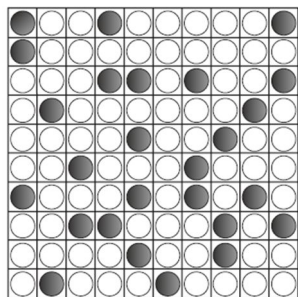


Crosslinked Polymers

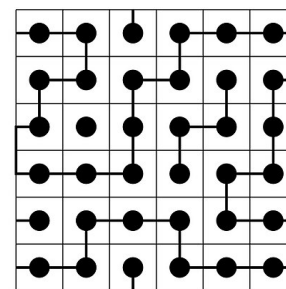
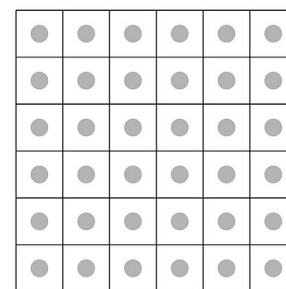
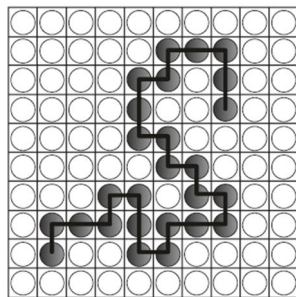
Solvent Quality (I)

The **Flory-Huggins lattice theory** defines solvent quality using the dimensionless interaction parameter (χ), which describes the energetic balance between polymer–solvent contacts and polymer–polymer or solvent–solvent contacts. It dictates whether polymer chains will swell to maximize solvent exposure or collapse to minimize it.

2 small molecules
(solvents)

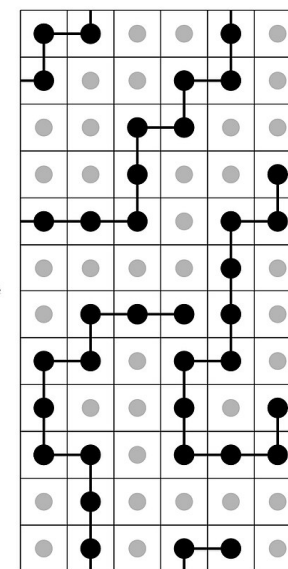


1 small molecule
+ 1 polymer



Solvent
+
Polymer

→ Mixture



Crosslinked Polymers

Solvent Quality (II)

The **Flory-Huggins theory** defines solvent quality using the dimensionless interaction parameter (χ), which describes the energetic balance between polymer–solvent contacts and polymer–polymer or solvent–solvent contacts. It dictates whether polymer chains will swell to maximize solvent exposure or collapse to minimize it.

$$\Delta G^m = RT \left(\underbrace{\varphi_1 \ln \varphi_1 + \frac{\varphi_2}{X_N} \ln \varphi_2}_{\text{Entropy term}} + \underbrace{\varphi_1 \varphi_2 \chi}_{\text{Enthalpy term}} \right)$$

https://en.wikipedia.org/wiki/Flory%E2%80%93Huggins_solution_theory

Crosslinked Polymers

Solvent Quality (III)

Solvent quality is classified by specific χ values and the macroscopic physical behavior of the polymer chain:

Good Solvent ($\chi < 0.5$)

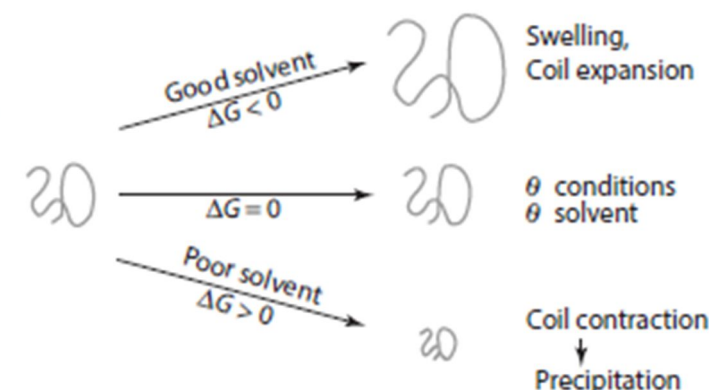
Polymer – solvent interactions are highly favorable. **Chains swell**, forming an extended, open coil structure to maximize contact with the solvent. The excluded volume is positive, meaning polymer segments repel one another.

Theta (Θ) Solvent ($\chi = 0.5$)

Favorable polymer–solvent contacts exactly balance the energetic penalties of polymer–polymer contacts (effectively, the second virial coefficient $A_2 = 0$). The polymer chain acts as an **ideal chain**, adopting its random unperturbed coil dimensions (pseudo-ideal conditions).

Poor Solvent ($\chi > 0.5$)

Polymer–solvent interactions are unfavorable. Polymer segments prefer to interact with each other rather than the solvent. **Chains collapse** into a compact, dense globule and can eventually **precipitate** out of solution.



Crosslinked Polymers

Solvent Quality (IV)

Additional factors

- Polyelectrolytes add Donnan osmotic pressure and counterion-specific effects.
- Temperature and salts (via both their nature and their concentration) can significantly impact swelling.
(see salting-out and -in effects)

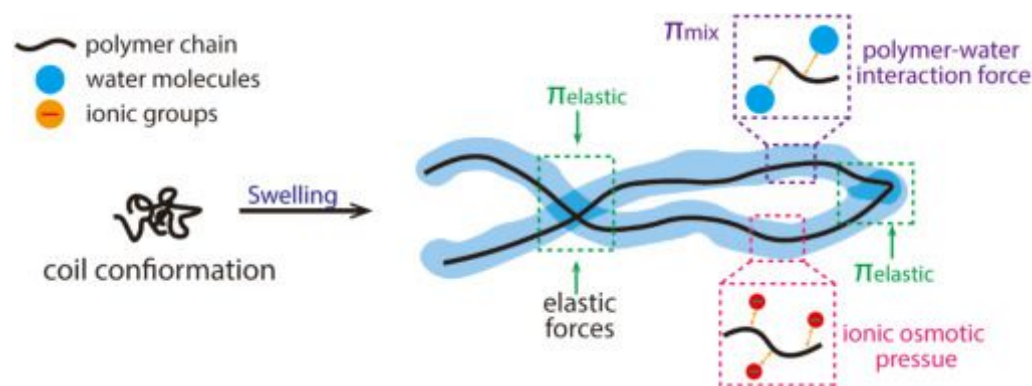
Crosslinked Polymers

Swelling Mechanisms (I)

Swelling in polymer networks occurs when a solvent penetrates the structure, causing the chains to solvate and expand the volume of the material.

The process is governed by a dynamic balance between two competing forces:

- **osmotic pressure**, which drives the solvent into the network;
- **elastic retraction**, which resists chain stretching.



Crosslinked Polymers

Swelling Mechanisms (II)

The network properties which govern the swelling process are:

- the crosslink density;
- the solvent quality;
- the entanglements.

Crosslinked Polymers

Flory–Rehner Theory (I)

Flory–Huggins asks:

"Will a polymer and solvent mix?,"

Flory–Rehner asks:

"Given that the polymer is permanently crosslinked and cannot dissolve, how much solvent will it absorb before mixing forces are balanced by network elasticity?"

Crosslinked Polymers

Flory–Rehner Theory (II)

Swelling occurs via (i) mixing (solvent quality, Flory–Huggins) and (ii) retraction (statistical theory of rubber elasticity). (iii) Equilibrium is reached when osmotic pressure is zero.

$$\pi_{\text{mix}} + \pi_{\text{el}} = 0.$$

$$\pi_{\text{mix}} = -\frac{RT}{v_1} \left[\ln(1 - \phi) + \phi + \chi \phi^2 \right]$$

$$\pi_{\text{el}} = \frac{RT}{v_1} \nu \left[\phi^{1/3} - \frac{\phi}{2} \right]$$

Crosslinked Polymers

Flory–Rehner Theory(III)

Swelling occurs via (i) mixing (solvent quality, Flory–Huggins) and (ii) retraction (statistical theory of rubber elasticity). (iii) Equilibrium is reached when osmotic pressure is zero.

$$\underbrace{\ln(1 - \phi) + \phi + \chi \phi^2}_{\pi_{\text{mix}}/(-RT/v_1)} + \underbrace{\nu \left(\phi^{1/3} - \frac{\phi}{2} \right)}_{\pi_{\text{el}}/(RT/v_1)} = 0$$

Flory–Rehner swelling equation

Crosslinked Polymers

Mesh Size (I)

The **Flory–Rehner swelling equation** (neutral gel, Eq. 8 in the previous answer) links the *equilibrium polymer volume fraction* ϕ to the **cross-link density** ν (number of elastically active network chains per unit dry volume).

By taking into account the polymer characteristics (e.g., dry density) and considering that a polymer strand is a random walk of N_e statistical segments (Kuhn segments) each of length b , the following is obtained:

$$\xi = b \sqrt{N_e} = b \sqrt{\frac{N_A}{\nu V_K}}.$$

Crosslinked Polymers

Mesh Size (II)

An equivalent form, where all quantities are measurable is:

$$\xi = b \sqrt{\frac{N_A}{V_K} \frac{\frac{\phi}{2} - \phi^{1/3}}{\ln(1 - \phi) + \phi + \chi\phi^2}}$$

- b Kuhn length (tables)
- N_A Avogadro number
- V_K molar volume of the Kuhn segment (often approximated as $V_K = b^3$ for flexible polymers, or taken from literature)
- ϕ equilibrium polymer volume fraction (obtained from swelling ratio Q ; $\phi \approx 1/Q$)

Q is determined during a swelling experiment, in which the swollen weight of a known weight of dry polymer is determined at a temperature for which the polymer bulk density and χ are known.

Crosslinked Polymers

Crosslink Density: Measurement (I)

Rubber Elasticity

A rubber (or any elastomeric polymer) consists of flexible macromolecular chains that are **permanently linked together at a finite number of points** (crosslinks).

Between the crosslinks the chains are free to adopt random-coil conformations, maximizing their (configurational) entropy.

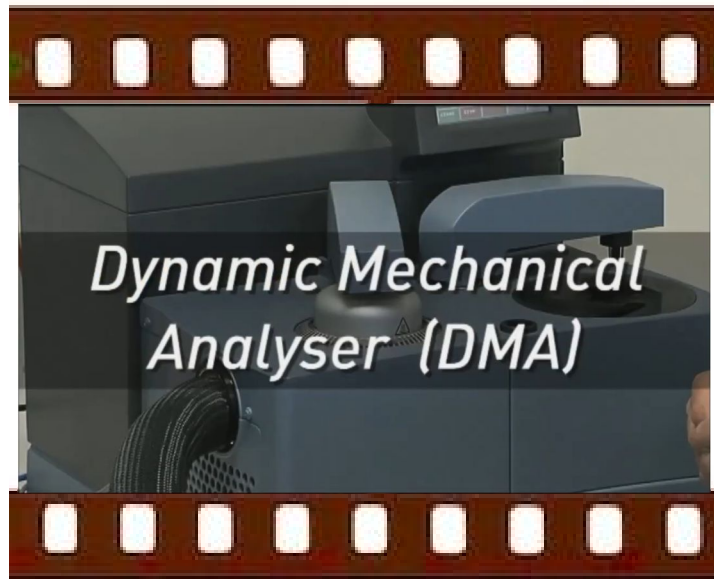
When the network is **deformed** (stretched, sheared, etc.), the chains are forced away from their random-coil statistics → **entropy is reduced**.

The shear modulus in the rubbery plateau: $G = \nu RT$

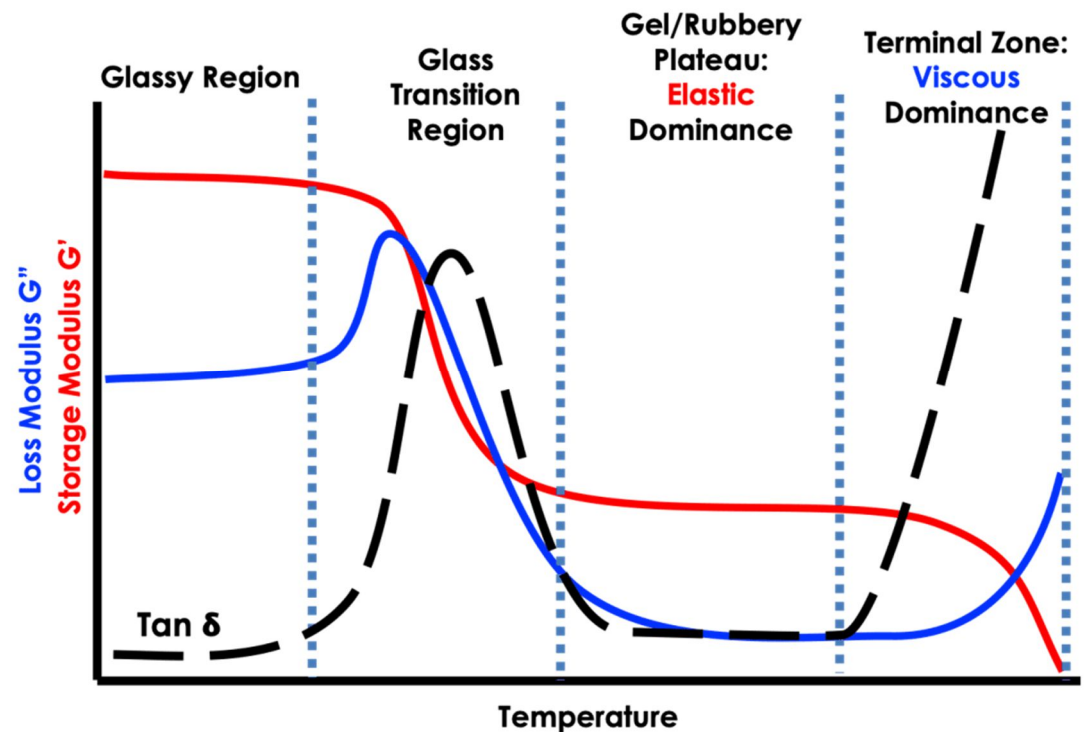
Crosslinked Polymers

Crosslink Density: Measurement (II)

Dynamic Mechanical Analysis



https://youtu.be/IxVeSTBP0cl?si=Y9D_LNpil8zDP7w0



<https://www.rheologylab.com/services/dynamic-mechanical-analysis-dma/>